

Workshop 14

Problem 1

Keywords: Substitution for Definite Integrals

Let f be a continuous function such that

$$\int_0^1 f(x)dx = 4, \quad \int_0^2 f(x)dx = -3, \quad \int_0^{-1} f(x^2)dx = 7.$$

Compute each of the following integrals, or explain why there is not enough information to do so.

(a) $\int_0^1 f\left(\frac{x}{2}\right) dx.$

(b) $\int_0^2 f\left(\frac{x}{2}\right) dx.$

(c) $\int_0^1 f(x^2) dx$

(d) $\int_0^1 xf(x^2) dx$

(e) $\int_{-1}^1 xf(x^2) dx$

(f) $\int_{-2}^2 xf(|x|) dx$

Problem 2

Keywords: Area Between Curves

Let R be the region in the xy -plane bounded between the lines $y = x$, $y = 2x$, and $x = 2$.

- (a) Sketch the region. Be sure to label the axes, the curves, the region R , and any points of intersection.
- (b) Set-up a definite integral (or a sum of definite integrals, if needed) using integration along the x -axis that computes the area of the region R .
- (c) Set-up a definite integral (or a sum of definite integrals, if needed) using integration along the y -axis that computes the area of the region R .
- (d) Evaluate one of your integrals (from parts (b) or (c)) to find the exact area of the region R .

Problem 3

Keywords: Area Between Curves, Unknown Constants

Let R be the region in the xy -plane bounded by the parabola $y = Cx^2$ and the line $y = C^2$, where C is a positive constant.

- (a) Sketch the region. Be sure to label the axes, the curves, the region R , and any points of intersection.
- (b) Set-up a definite integral (or a sum of definite integrals, if needed) using integration along the x -axis that computes the area of the region R .
- (c) Set-up a definite integral (or a sum of definite integrals, if needed) using integration along the y -axis that computes the area of the region R .
- (d) The area of the region R is $\frac{128}{3}$. Find the value of the constant C .

Extra Problem (*for practice, no need to submit*)

Let $f(x)$ be a continuous odd function, and let $g(x)$ be a continuous even function. Determine the parity of each of the following functions.

(a) $A(x) = \int_0^x f(t)dt.$

(b) $B(x) = \int_0^x g(t)dt.$

(c) $C(x) = \int_0^{f(x)} f(t)dt.$

(d) $D(x) = \int_0^{g(x)} f(t)dt.$